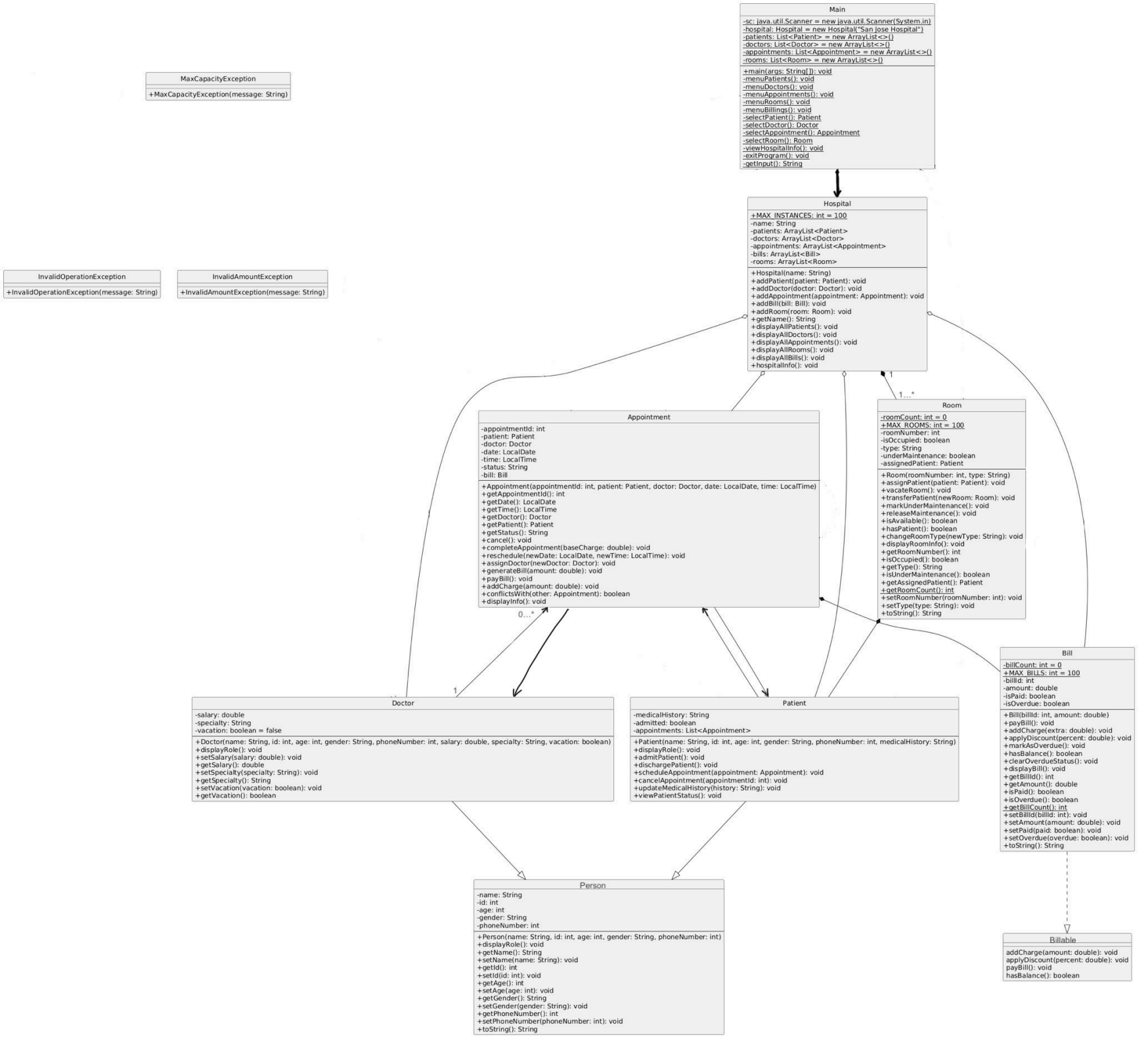
Main
-sc: java.util.Scanner = new java.util.Scanner(System.in)
-hospital: Hospital = new Hospital("San Jose Hospital")
-patients: List<Patient> = new ArrayList<>()
-doctors: List<Doctor> = new ArrayList<>()
-appointments: List<Appointment> = new ArrayList<>()
-rooms: List<Room> = new ArrayList<>()
+main(args: String[]): void
-menuPatients(): void
-menuDoctors(): void
-menuAppointments(): void
-menuRooms(): void
-menuBillings(): void
-selectPatient(): Patient
-selectDoctor(): Doctor
-selectAppointment(): Appointment
-selectRoom(): Room
-viewHospitalInfo(): void
-exitProgram(): void
-getInput(): String
MaxCapacityException
+MaxCapacityException(message: String)
Hospital
+MAX_INSTANCES: int = 100
-name: String
-patients: ArrayList<Patient>
-doctors: ArrayList<Doctor>
-appointments: ArrayList<Appointment>
-bills: ArrayList<Bill>
-rooms: ArrayList<Room>
+Hospital(name: String)
+addPatient(patient: Patient): void
+addDoctor(doctor: Doctor): void
+addAppointment(appointment: Appointment): void
+addBill(bill: Bill): void
+addRoom(room: Room): void
+getName(): String
+displayAllPatients(): void
+displayAllDoctors(): void
+displayAllAppointments(): void
+displayAllRooms(): void
+displayAllBills(): void
+hospitalInfo(): void
InvalidOperationException
+InvalidOperationException(message: String)
InvalidAmountException
+InvalidAmountException(message: String)
1
1...*
Room
-roomCount: int = 0
+MAX_ROOMS: int = 100
-roomNumber: int
-isOccupied: boolean
-type: String
-underMaintenance: boolean
-assignedPatient: Patient
+Room(roomNumber: int, type: String)
+assignPatient(patient: Patient): void
+vacateRoom(): void
+transferPatient(newRoom: Room): void
+markUnderMaintenance(): void
+releaseMaintenance(): void
+isAvailable(): boolean
+hasPatient(): boolean
+changeRoomType(newType: String): void
+displayRoomInfo(): void
+getRoomNumber(): int
+isOccupied(): boolean
+getType(): String
+isUnderMaintenance(): boolean
+getAssignedPatient(): Patient
+getRoomCount(): int
+setRoomNumber(roomNumber: int): void
+setType(type: String): void
+toString(): String
Appointment
-appointmentId: int
-patient: Patient
-doctor: Doctor
-date: LocalDate
-time: LocalTime
-status: String
-bill: Bill
+Appointment(appointmentId: int, patient: Patient, doctor: Doctor, date: LocalDate, time: LocalTime)
+getAppointmentId(): int
+getDate(): LocalDate
+getTime(): LocalTime
+getDoctor(): Doctor
+getPatient(): Patient
+getStatus(): String
+cancel(): void
+completeAppointment(baseCharge: double): void
+reschedule(newDate: LocalDate, newTime: LocalTime): void
+assignDoctor(newDoctor: Doctor): void
+generateBill(amount: double): void
+payBill(): void
+addCharge(amount: double): void
+conflictsWith(other: Appointment): boolean
+displayInfo(): void
0...*
1
Bill
-billCount: int = 0
+MAX_BILLS: int = 100
-billId: int
-amount: double
-isPaid: boolean
-isOverdue: boolean
+Bill(billId: int, amount: double)
+payBill(): void
+addCharge(extra: double): void
+applyDiscount(percent: double): void
+markAsOverdue(): void
+hasBalance(): boolean
+clearOverdueStatus(): void
+displayBill(): void
+getBillId(): int
+getAmount(): double
+isPaid(): boolean
+isOverdue(): boolean
+getBillCount(): int
+setBillId(billId: int): void
+setAmount(amount: double): void
+setPaid(paid: boolean): void
+setOverdue(overdue: boolean): void
+toString(): String
Doctor
-salary: double
-specialty: String
-vacation: boolean = false
+Doctor(name: String, id: int, age: int, gender: String, phoneNumber: int, salary: double, specialty: String, vacation: boolean)
+displayRole(): void
+setSalary(salary: double): void
+getSalary(): double
+setSpecialty(specialty: String): void
+getSpecialty(): String
+setVacation(vacation: boolean): void
+getVacation(): boolean
Patient
-medicalHistory: String
-admitted: boolean
-appointments: List<Appointment>
+Patient(name: String, id: int, age: int, gender: String, phoneNumber: int, medicalHistory: String)
+displayRole(): void
+admitPatient(): void
+dischargePatient(): void
+scheduleAppointment(appointment: Appointment): void
+cancelAppointment(appointmentId: int): void
+updateMedicalHistory(history: String): void
+viewPatientStatus(): void
Person
-name: String
-id: int
-age: int
-gender: String
-phoneNumber: int
+Person(name: String, id: int, age: int, gender: String, phoneNumber: int)
+displayRole(): void
+getName(): String
+setName(name: String): void
+getId(): int
+setId(id: int): void
+getAge(): int
+setAge(age: int): void
+getGender(): String
+setGender(gender: String): void
+getPhoneNumber(): int
+setPhoneNumber(phoneNumber: int): void
+toString(): String
Billable
addCharge(amount: double): void
applyDiscount(percent: double): void
payBill(): void
hasBalance(): boolean